DEV DELVADIYA (DEV PATEL)

Undergraduate Researcher · Computer Vision, Vision-Language Models & 3D Medical Vision
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EDUCATION

Gujarat Technological University — Ahmedabad, Gujarat, India
Bachelor of Engineering in Computer Science (B.E. CS '27) | ***CGPA: 9.35 / 10.0*** *(Expected June 2027)*

Government Polytechnic Rajkot — Rajkot, Gujarat, India
Diploma in Information Technology | ***CGPA: 9.32 / 10.0*** *(2021 — 2024)*

RESEARCH APPOINTMENTS

Xu Lab, Carnegie Mellon University — Remote
Research Intern | Advisor: Prof. Min Xu | Feb 2026 — Present

- Developing deep learning non-rigid registration pipelines for 3D cortical surface meshes derived from raw T1-weighted brain MRI scans.
- Interfacing FreeSurfer (recon-all) pipelines with Freeview to maintain spherical conformal mappings, sulcal depth continuity, and topological surface guarantees.
- Distributed a fully preprocessed IXI benchmark subject in a public repository to lower computational barriers for 3D medical researchers.

Gujarat Technological University — Ahmedabad, India
Undergraduate Researcher | Advisor: Prof. Raj Hakani | Oct 2025 — Present

- Leading design of **MedOptm**, a compact clinical Vision-Language Model pairing CvT-24 with a 1.7B LLaMA backbone.
- Designed an architectural Token Packer achieving 75% token compression, yielding 0.33 METEOR on clinical VQA under consumer GPU constraints.

PUBLICATIONS

Two-Stage Transfer Learning with Modified VGG16 for Multi-Class Brain Tumor Classification on Heterogeneous MRI
Dev Delvadiya, *Raj Hakani*
International Conference on Machine Learning and Data Engineering (ICMLDE 2025). Accepted for presentation.

- Devised a two-stage training strategy freezing feature extractors before unfreezing upper convolutional blocks, achieving 97.88% test accuracy across 4 tumor classes.

HONORS & COMPETITIVE PROGRAMS

Amazon ML Summer School 2026 — Selected Participant (Sixth Edition)

- Selected through a competitive national screening test covering multivariate calculus, probability distributions, ML algorithms, and coding.
- Intensive training modules led by Amazon Applied Scientists on multimodal deep learning, distributed training, and recommendation systems.

SELECTED RESEARCH PROJECTS

CvT-24: Convolutional Vision Transformer from First Principles | PyTorch, Custom Attention

- Implemented Microsoft's CvT-24 entirely from scratch in PyTorch with custom depthwise separable convolutional token projections and hierarchical stages without library wrappers.

Fashion Multimodal Semantic Retrieval | FashionCLIP, FAISS, Contrastive Adapters

- Built fine-grained cross-modal search engine solving relational attribute binding collapse in frozen CLIP encoders with sub-0.8ms query latency.

BrainScan AI: Multi-Class MRI Diagnostic Interface | PyTorch, Streamlit, ICMLDE '25 Model

- Deployed clinical diagnostic web application for real-time brain MRI classification with confidence distribution and visual report generation.

TECHNICAL SKILLS

Languages & Core: Python, C, CUDA C++, C#, SQL, Bash, Git, LaTeX

Frameworks & Libraries: PyTorch, Torchvision, Scikit-Learn, OpenCV, NumPy, Pandas, FlashAttention-2, FAISS
Medical & 3D Imaging: FreeSurfer (recon-all), Freeview, PyTorch3D, MeshLab, NIfTI (.nii/.mgz), 3D Cortical Surface Registration
Deployment & Tools: Streamlit, Hugging Face Hub, Docker, Linux Workstations